

Joonyoung KIM

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EDUCATION

Ajou University

Expected Graduation Date: February 2027

GPA: 4.05/4.5

B.S. in Transportation System Engineering, Minor in Future Car

PUBLICATION

Kim, J. Y., Kim, G. H., Kim, D. C., An, H. S., Kang, T. S., & Kim, E. J. (2026). A Hierarchical Bayesian Model for Estimating Nationwide Demand for Demand-Responsive Transit at a Fine Spatial Resolution. *Journal of Korean Society of Transportation*, Accepted.

RESEARCH EXPERIENCE

Demand Forecasting Model for Demand-Responsive Transit

Paran Semester, Industry-Proposed Project | Sep. 2025 – Dec. 2025

- Developed a demand forecasting model for demand-responsive transit using Hierarchical Bayesian gamma regression
- Manuscript accepted for publication in the Journal of Korean Society of Transportation

Correlation Analysis between Micromobility and Conventional Public Transit

Undergraduate course 'Traffic Demand Forecasting Project' | Mar. 2025 – Jun. 2025

- Applied Bayesian structural time-series causal impact analysis to examine the relationship between micromobility usage and public transit ridership
- Interpreted the results in relation to land-use characteristics and public transit accessibility

Suwon Smart City Project

Undergraduate course 'Smart Mobility Project' | Sep. 2024 – Dec. 2024

- Analyzed using SUMO simulator
- Proposed traffic mitigation strategies for congestion on Maesan-ro in Suwon

SKILLS

Software Tool Skills: Python(PyTorch, Pandas), SUMO, VISSIM

Language: Korean(native), English(Advanced)

SCHOLARSHIPS

Full-Tuition Merit Scholarship, Ajou University | 2021–Present

- Awarded as the top-ranked admitted student.

Dean's Recommendation Scholarship, Graduate School of Transportation and Mobility, Ajou University | Fall 2025 – Spring 2026

- Awarded to an undergraduate student upon the Dean's recommendation.